



Implementation of an abduction algorithm based on MARCO in the CATS solver

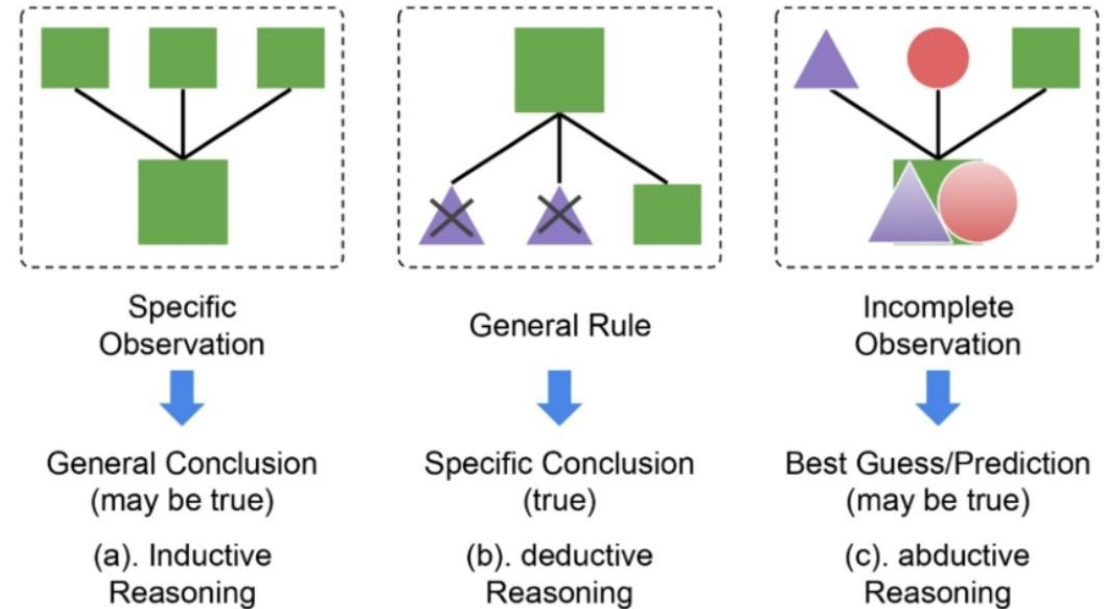
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Abduction

- ▶ A form of reasoning which generates plausible explanations for given observations
- ▶ To solve abductive problems, we need to have a knowledge base, an observation and a set of abducibles
- ▶ The knowledge base plus the explanation must entail the observation: $K \cup E \models O$
- ▶ This is logically equivalent to: $K \cup E \cup \{\neg O\}$ is inconsistent



Traditional abduction algorithms

- ▶ e.g. MHS, MXP, HST
 - ▶ Search for models that satisfy $K \cup E$
 - ▶ Verify that these models entail the observation
 - ▶ They are model-based → slow as the knowledge base grows
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- ▶ HS-tree-based methods require model computation, which is not a commonly provided service by reasoners
 - ▶ We want to explore an alternative approach inspired by the MARCO algorithm that does not use models

MARCO algorithm

- ▶ An algorithm for exploring the structure of constraint systems by enumerating MUSes (Minimal Unsatisfiable Subsets) and MSSes (Maximal Satisfiable Subsets)
 - ▶ Pick a candidate subset of constraints
 - ▶ Check if it's satisfiable
 - ▶ If unsatisfiable → extract a MUS
 - ▶ If satisfiable → expand to an MSS
 - ▶ Block that region and continue
- ▶ It uses a systematic exploration using only satisfiability checks, not model enumeration

CATS solver

- ▶ Comenius-Abduction-Team Solver
- ▶ An experimental solver for abductive reasoning generating explanations for observed facts
- ▶ Given a background theory, a set of observations, and a list of abducibles, CATS finds minimal, consistent hypotheses that explain the observations
- ▶ Offers switches and different optimisations
- ▶ Algorithms: HST, MHS, MHS-MXP, MXP, RCT, HST-MXP, RCT-MXP, QXP



Main goals of this thesis

- ▶ Analysis of the state of the art, including the existing abduction tool CATS, the principles of HS-tree-based methods, and the incomplete MergeXplain method.
- ▶ Design and modular implementation of an abduction algorithm inspired by the MARCO algorithm and incorporating the MergeXplain method.
- ▶ Optimisation of the implemented abduction algorithm.
- ▶ Empirical evaluation of the implemented algorithm and its comparison with HS-tree-based methods.

Studying literature

- ▶ Description logics
 - ▶ S. Rudolph: Foundations of description logics, 2011
 - ▶ F. Baader, I. Horrocks, U. Sattler: Description logics, 2008
 - ▶ F. Baader, I. Horrocks, C. Lutz, U. Sattler. An Introduction to Description Logic, 2017
- ▶ Abduction
 - ▶ C. Elsenbroich, O. Kutz, U. Sattler: **A Case for Abductive Reasoning over Ontologies**, 2006
- ▶ Abduction algorithms
 - ▶ J. Pukancová, M. Homola: Tableau-Based ABox Abduction for the ALCHO Description Logic, 2017
 - ▶ M. Homola, J. Pukancová, J. Boborová, I. Balintová: Merge, Explain, Iterate: A Combination of MHS and MXP in an ABox Abduction Solver
- ▶ MARCO algorithm
 - ▶ M. H. Liffiton, A. Previti, A. Malik, J. Marques-Silva: **Fast, flexible MUS enumeration**, 2016

Implementation of MARCO for abduction

- ▶ Getting to know the CATS solver
- ▶ Integrating the new algorithm into CATS – creating new classes, switches in Java
- ▶ First version is using straightforward enumeration
- ▶ Removing duplicities, filtering non-minimal explanations

Implementation of MARCO for abduction

- ▶ More effective version is using bitmaps to generate subsets
- ▶ Checking consistent supersets and inconsistent subsets to reduce the number of consistency checks
- ▶ Using type BitSet
- ▶ Testing on various inputs and creating unit tests
- ▶ Experimenting with checking subsets in different orders (from biggest to smallest, randomly, MXP-like)

Planned tasks

- ▶ Designing and implementing various versions of the algorithm
- ▶ Testing the versions of the algorithm on multiple inputs
- ▶ Evaluate the performance of the algorithm
- ▶ Compare the performance to other methods

Sources

- ▶ https://www.researchgate.net/figure/Simple-illustrations-of-three-types-of-reasoning_fig1_387975102
- ▶ <https://ceur-ws.org/Vol-4091/paper34.pdf>